

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

C02: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I Deepan Kumar P (192312498) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

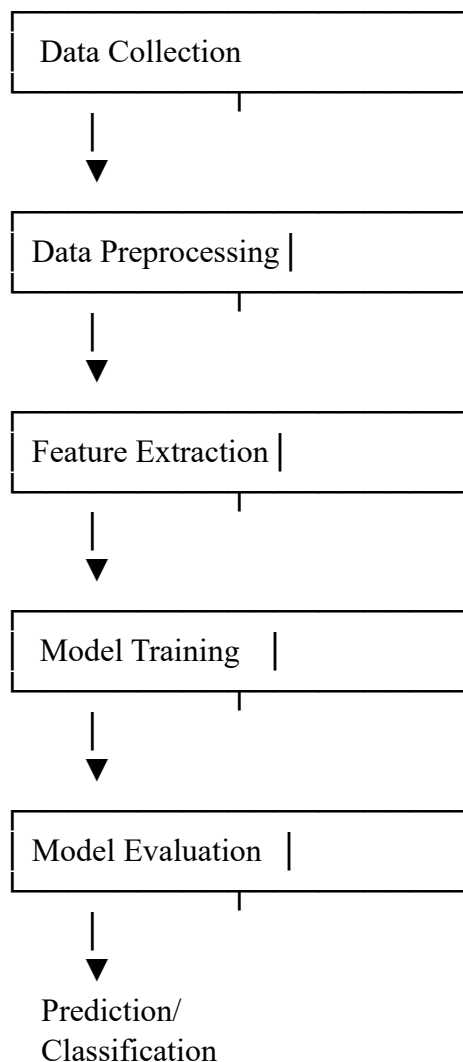
Deepan Kumar P

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

Introduction

A **Machine Learning (ML) system** is a sequence of processes that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed. The workflow begins with collecting data, preparing it for analysis, extracting useful information, training a model, and finally evaluating its performance.

MACHINE LEARNING SYSTEM



Explanation of Functional Blocks

1. Data Collection:

Data is collected from sources such as databases, sensors, websites, or IoT devices. The quality of data affects the model's performance.

2. Data Preprocessing:

The collected data is cleaned by removing missing values, duplicates, and noise. This improves the accuracy of the model.

3. Feature Extraction:

Important features are selected from the dataset to reduce complexity and improve learning.

4. Model Training:

The processed data is used to train a machine learning algorithm so that it learns patterns from the data.

5. Model Evaluation:

The trained model is tested using new data to measure its performance using metrics such as accuracy.

Examples

Supervised Learning (Labelled Data):

Example: Student Pass/Fail Prediction

The dataset contains study hours and result (Pass/Fail). Since the correct output (label) is available, the model learns to predict whether a new student will pass or fail.

Unsupervised Learning (Unlabelled Data):

Example: Customer Segmentation

The dataset contains only customer information (age, income, etc.) without labels. The algorithm groups similar customers into clusters based on their characteristics.

Conclusion:

A machine learning system consists of data collection, preprocessing, feature extraction, model training, and evaluation. Supervised learning uses labelled data for prediction, while unsupervised learning uses unlabelled data to identify patterns or clusters.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

Historical Data + Real-time Data



Data Preprocessing



Feature Selection



Model Training



Validation & Evaluation



Flood/Earthquake Prediction



Early Warning & Emergency Response

A Machine Learning (ML) system for disaster management uses historical data (past flood records, earthquake history, rainfall data) and real-time data (weather stations, satellite images, river water levels, and IoT sensors) to predict disasters and provide timely alerts.

Model Selection

A **Neural Network (Artificial Neural Network - ANN)** is suitable because it can identify complex patterns from large datasets and produce accurate predictions. It can effectively process both historical and real-time information, making it ideal for disaster prediction.

Data Preprocessing and Feature Selection

Before training the model, the collected data is pre-processed by:

- Removing missing and duplicate values.
- Normalizing numerical data.
- Eliminating noisy or inconsistent data.

Important features selected for prediction include:

- Rainfall intensity
- River water level
- Soil moisture
- Temperature
- Seismic activity
- Historical disaster records

Selecting relevant features improves prediction accuracy and reduces training time.

Training, Validation, and Evaluation

The model is trained using historical disaster data. It is then validated using a separate validation dataset to fine-tune the model. Finally, the model is evaluated using performance metrics such as **Accuracy, Precision, Recall, and F1-Score** to ensure reliable predictions.

Applications

The proposed ML system helps authorities by:

- Providing early warning alerts.
- Identifying high-risk areas.
- Supporting risk assessment.
- Assisting in evacuation planning and emergency response.
- Reducing loss of life and property.

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

A Multilayer Perceptron (MLP) is a type of neural network used for classification and prediction tasks. For binary classification, the output is either 0 or 1. During training, the backpropagation algorithm updates the weights of the network to reduce prediction errors.

Working of Backpropagation

1. Forward Pass:

The input data is passed through the hidden layer to the output layer. Each neuron calculates a weighted sum and applies an activation function to produce the predicted output.

Example:

Suppose the actual output is 1, but the network predicts 0.8.

2. Error Calculation:

The prediction error is calculated by comparing the predicted output with the actual output.

Error = Actual Output – Predicted Output

$$\text{Error} = 1 - 0.8 = 0.2$$

3. Backward Propagation:

The calculated error is propagated backward through the network. The weights are adjusted using gradient descent so that the error is reduced in the next iteration. This process is repeated until the model achieves good accuracy.

Role of Learning Rate

The learning rate determines how much the weights are updated during each iteration.

- A high learning rate speeds up training but may skip the optimal solution.
- A low learning rate provides better accuracy but increases training time.

Hence, selecting an appropriate learning rate ensures faster convergence and stable learning.

Role of Activation Functions

Activation functions introduce non-linearity, enabling the network to learn complex patterns.

Common activation functions include:

- Sigmoid – Used for binary classification.
- ReLU (Rectified Linear Unit) – Commonly used in hidden layers for faster learning.

- Tanh – Produces outputs between -1 and 1.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyse how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

A Genetic Algorithm (GA) is an optimization technique inspired by natural evolution. It improves candidate solutions over multiple generations using selection, crossover, and mutation. In this problem, the objective is to maximize the fitness value based on Speed and Accuracy.

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Step-by-Step Genetic Algorithm Process

1. Initial Population:

All five examples are considered as the initial population. Their fitness values range from **40 to 80**.

2. Selection:

Individuals having higher fitness values have a greater chance of being selected as parents. Therefore, Example 5 (80) and Example 3 (75) are selected as strong parent solutions.

3. Crossover:

Features of selected parents are combined to produce new offspring. For example, crossover between (Medium Speed, High Accuracy) and (High Speed, Medium Accuracy) may produce (High Speed, High Accuracy), which is a promising new candidate.

4. Mutation:

Mutation randomly changes a feature, such as changing Speed from Medium to High. It introduces diversity and helps the algorithm explore new solutions instead of getting stuck at a local optimum.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks) Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

A Neural Network is trained using the Backpropagation Algorithm to classify whether a patient has a disease based on Age, Blood Pressure, Glucose Level, and Lifestyle Risk.

Working of Backpropagation

1. Forward Pass:

The input features are passed through the hidden layer to generate the predicted output.

2. Error Calculation:

The predicted output is compared with the actual output, and the prediction error is calculated.

3. Backpropagation:

The error is propagated backward through the network, and the weights are updated using gradient descent. This process is repeated until the prediction error is minimized.

Actual / Predicted	Yes	No
Yes	TP = 2	FN = 1
No	FP = 1	TN = 1

Therefore,

- **True Positive (TP) = 2**
- **True Negative (TN) = 1**
- **False Positive (FP) = 1**
- **False Negative (FN) = 1**

Performance Evaluation

Accuracy

$$= (2 + 1) / 5$$

$$= 3/5 = 0.60 = 60\%$$

Precision

$$= 2 / (2 + 1)$$

$$= 66.67\%$$

Recall

$$= 2 / (2 + 1)$$

$$= 66.67\%$$

The model correctly predicts **3 out of 5** cases, resulting in an **accuracy of 60%**.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand